

Advanced Networked Systems Security (ANSS) EP2510 / FEP3300

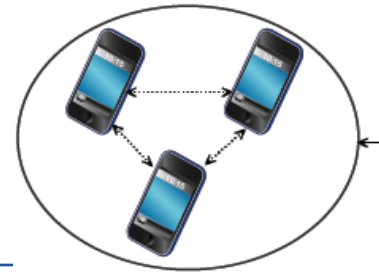
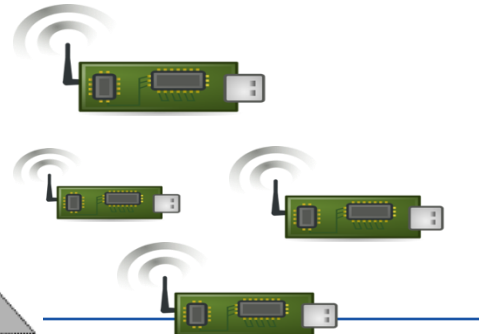
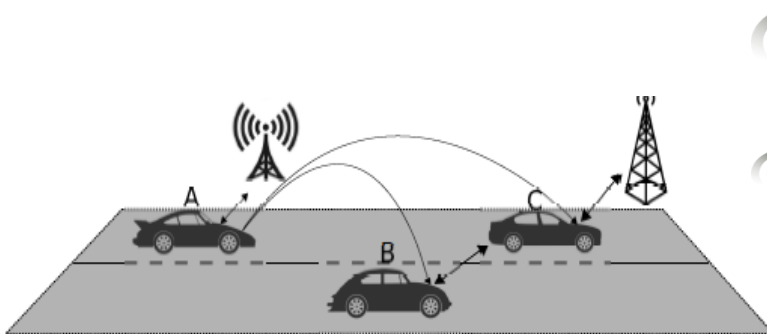
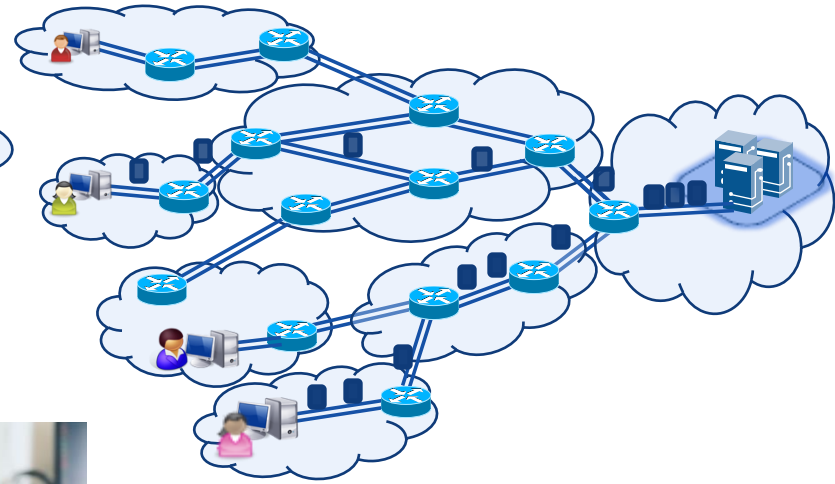
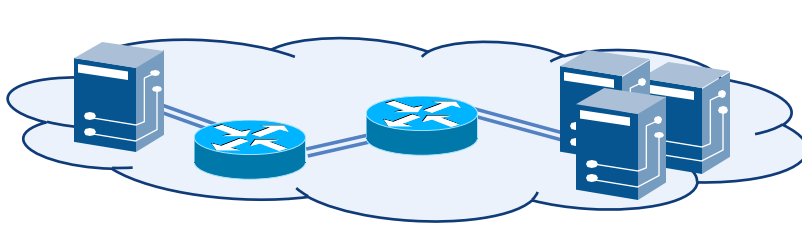
Fall 2025



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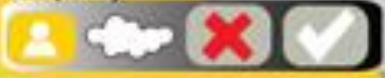
street faults



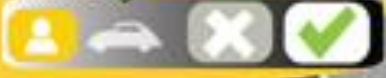
street faults



air quality



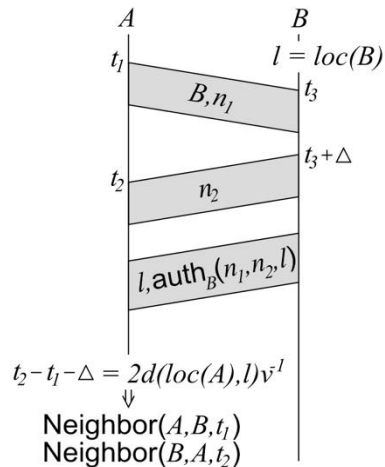
carsharing



Are things going to get better or worse?

Smart environments **could become horror environments unless** we design security (and privacy) correctly

Lessons learned

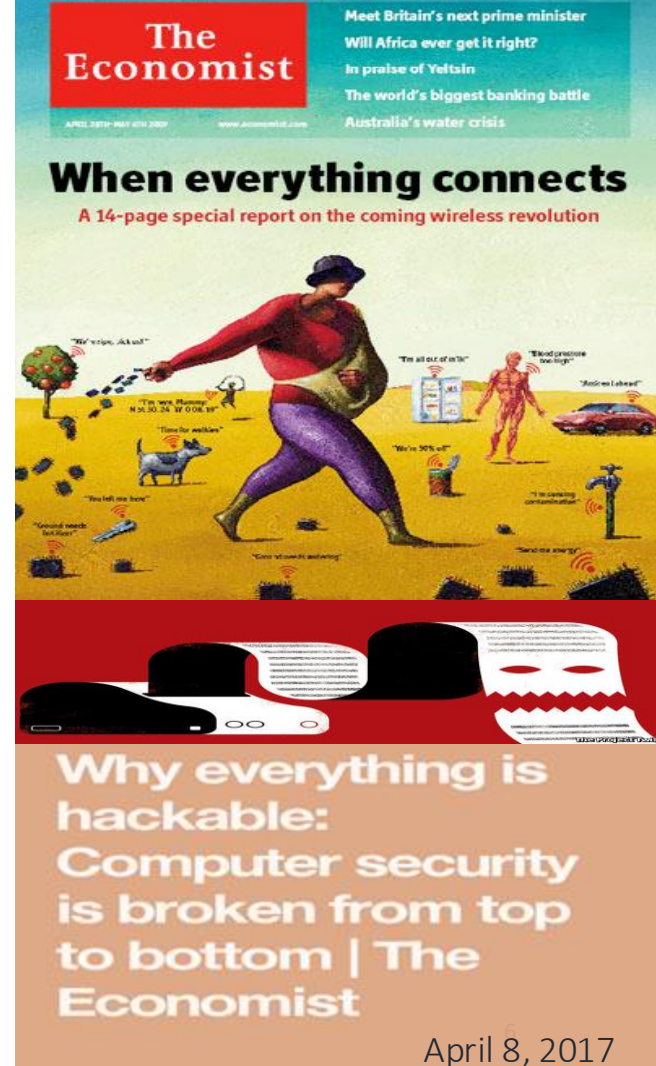


PRESERVE
preparing secure V2X communication systems

Final event, 2015

CAR 2 CAR
COMMUNICATION CONSORTIUM

First demo, 2008





Networked Systems Security



Networked Systems Security (NSS)

- How to secure networked systems?
- Emphasis on concepts across technologies
- Contemporary and emerging technologies

NSS (cont'd)

- Attacks - adversarial behavior
 - Adversary
 - > *Has/introduces own devices/nodes in the system*
 - > *Compromises devices/nodes already in the system*
 - Adversarial devices
 - > *Do not do what they were supposed to do*
 - > *Do not do what benign, correct devices do*

Disruption, manipulation, destruction: Unacceptable!

Prevent and manage adversarial behavior!



NSS (cont'd)

- Decide 'who' can trust 'who'
- Design new protocols
- Re-design your existing protocols
 - Add security
 - Possibly disable functionality
- Make sure that the secure(d) protocols work
 - Do applications still do what they are supposed to?
 - Why? Because security can always add complexity
- Prove they are secure



Building Networked Systems Security (BNSS)

- All the above steps to design a secure solution

AND

- Implement your solutions
- Verify
- Evaluate



BNSS (cont'd)

- Wireless and wire-line security
- Real-world technologies
- Real-world problems
- Real-world solutions
 - Implemented and demonstrated by you



Advanced Networked Systems Security (ANSS)

- Networked systems deployed
- Security solutions deployed
 - Some good, some bad...
- Systems keep changing
- Security is a buzz-word, BUT it can
 - Be under-estimated
 - Considered a nuisance
 - Over-sold



ANSS (cont'd)

- Networked systems keep changing
 - Growing larger
 - Devices getting smaller
 - New communication links
 - New applications
 - New requirements
- New security and privacy solutions needed
- How? Research!



ANSS (cont'd)

- Large community
 - People write papers
 - Meet in conferences
 - Publish in journals
- Academic and industry crowds
- Universities
- Industry research labs
- Government labs, organizations



ANSS (cont'd)

- Research?
 - Looking at new questions
 - > *How new?*
 - Well-defined technical problem
 - State-of-the-art
 - Novel approach to problem
 - > *New and better solution*
 - > *Analysis of a system*
 - > *Systematic comparison*
 - Or new, never thought-of before, interesting problem
 - > *And convincing solution*



ANSS (cont'd)

- We dive into the latest and hottest
- Read, present, and debate on pros and cons
- Choose one topic and go deeper
 - Critically survey what is done
 - Identify what is not satisfactorily or at all addressed
 - Identify an interesting question/problem
 - Reflect and write about it
 - Craft your solution to it

Caution: Given the course dimensions, adjust the scope



ANSS (cont'd)

- Course format
 - 5 modules
 - > *Presented material in seminar form @ meeting i*
 - > *Reading list of two or three papers*
 - > *Discussion @ meeting i+1*
 - > *Submit on canvas answers, after the discussion, to questions based on the reading list*
 - Simple, brief writing
 - Submission: Wed after the discussion, AoE
 - > *Possibly a 6th module this year!*
 - Individual project
 - Mid-term/proposal presentations
 - Final presentations
 - Final reports



ANSS (cont'd)

- Schedule
 - November 4, 2025, U614 @ 13:00
 - > **Seminar: Vehicular system security and privacy (VCS S&P)**
 - > **No discussion**
 - > *Reading list discussion leads at the next meeting: Kostas and Sheng*
 - November 10, 2025, LV3 Henrik Eriksson @ 13:00
 - > **Seminar: Secure Localization and Positioning (SecPos)**
 - > **Discussion on VCS S&P**
 - > *Reading list discussion leads at the next meeting: Andrew and Kostas*
 - November 19, 2025, LV3 Henrik Eriksson @ 13:00
 - > **Seminar: Secure networked sensing (SecSens)**
 - > **Discussion on SecPos**
 - > *Reading list discussion leads at the next meeting: Zahra and Sheng*



ANSS (cont'd)

- Schedule
 - November 25, 2025, K53 @ 13:00
 - > **Seminar: Privacy enhancing technologies for mobile computing (PETSmob)**
 - > **Discussion on SecSens**
 - > *Reading list discussion leads at the next meeting: Marco and Ahmed*
 - December 5, 2025, LV3 Henrik Eriksson @ 13:00
 - > **Seminar: Secure Routing protocols (SRPs)**
 - > **Discussion on PETSmob**
 - > *Reading list discussion leads at the next meeting: Ahmed and Marco*
 - December 9, 2025, online @ 10am
 - > **Recap / review / Q/A & Discussion on SRPs**
 - December 11 and December 12, 2025, M24 and H1 @ 09:00 and 11:00
 - > **Mid-term project presentations**
 - > *Schedule to be settled based on the final number of projects and individual constraints*



ANSS (cont'd)

- January 8, 9, 2026 @ V11, V12, 14:00
 - **Final presentations workshop**
 - Schedule to be settled based on the final number of projects and individual constraints
 - Draft report and slides submitted the same day



ANSS (cont'd)

- Project (individual or team of 2)
 - Research oriented, hands-on or theoretical
 - Work on a project closely related to one of the modules
 - Topic proposed and refined with the help of NSS
 - Work closely with two members of the teaching team
- Project milestones/submissions to canvas
 - Project group formation: Nov. 15, 2025
 - Topic selection & abstract: Nov. 24, 2025
 - One page abstract: Dec. 8, 2025
 - Mid-term presentation: Dec. 11 or 12, 2025
 - Interim report: Dec. 19, 2025
 - Peer review on interim report: Dec. 22, 2025
 - Draft final report and slides: January 7, 2026 (**12:00 noon**)
 - Peer review on draft final report: January 9, 2026 (AoE)
 - Finalized report: January 31, 2026

Secure and privacy-preserving vehicular communications

- Secure vehicle-to-vehicle and vehicle-to-infrastructure protocols
- Privacy enhancing technologies
- Thwarting clogging denial of service attacks
- Security and privacy for car sharing services
- Secure driving automation – platooning

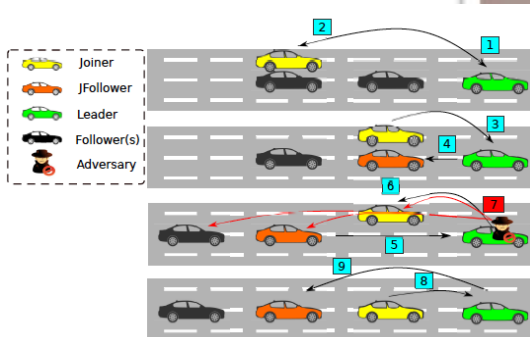
Standardization & harmonization
(IEEE 1609.2, ETSI, C2C-CC)



Final event, 2015



First demo, 2008



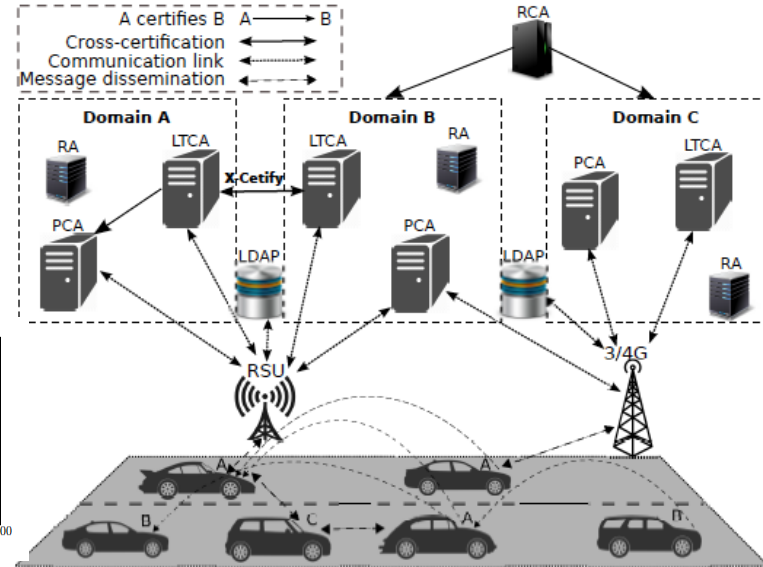
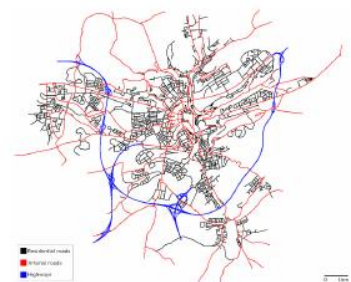
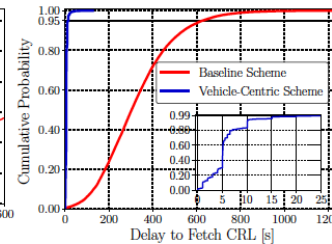
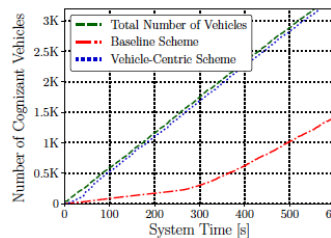
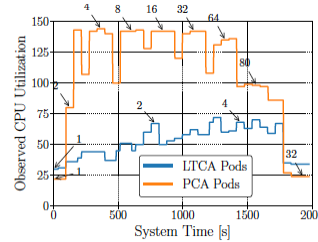
[ACM WiSec 2022, ACM WiSec 2019, Ad hoc Networks 2019, IEEE TMC 2021, IEEE IoT-J 2021]

Identity and credential management for large scale mobile systems

- Management of credentials: provisioning, revocation, resolution
- Challenges
 - Global deployment
 - Very large numbers of credentials (five orders of magnitude more)
 - Dynamically changing loads
 - Balancing act: security, privacy, and efficiency
- Scalable cloud-based system
- Efficient revocation

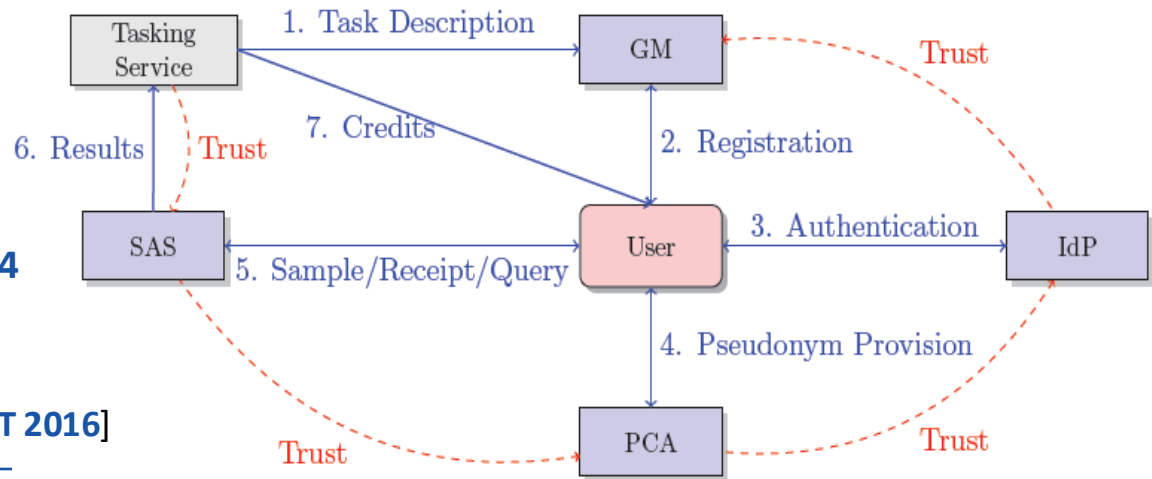
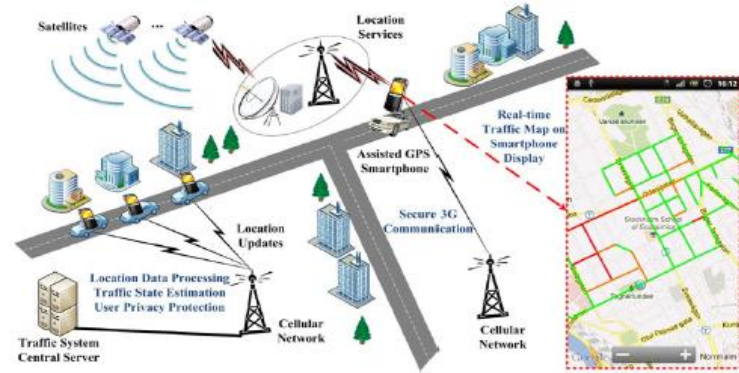
Internet Society/IRTF Applied Networking Research Prize 2018

[IEEE ITS 2018, ACM WiSec 2018, ACM WiSec 2019]



Identity and credential management for large scale mobile systems (cont'd)

- Protect users from the system (privacy)
 - Anonymity (conditional), unlinkability
- Protect the system from the users (security)
 - Authentication, authorization
 - Accountability
 - Misbehavior detection
- User incentives



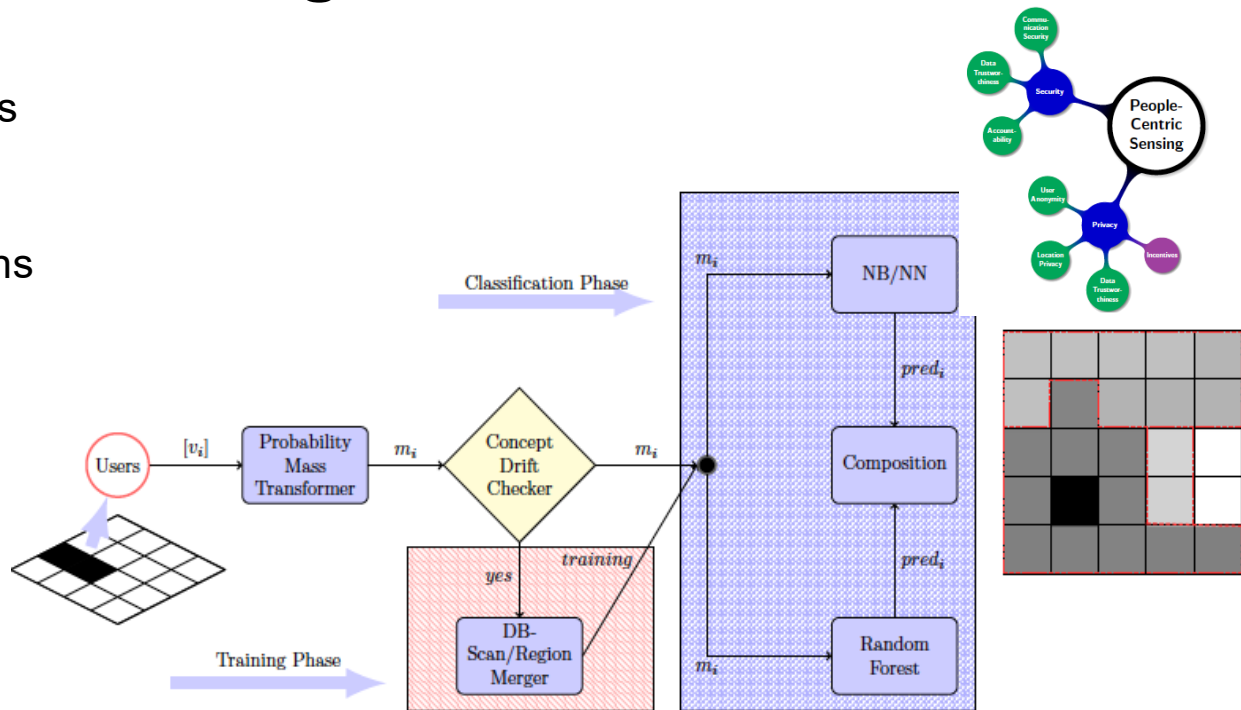
ACM WiSec Best Paper Award 2014

[ACM WiSec 2014, IEEE ITS 2015, IEEE IoT 2016]

Mobile crowd-sensing: Data Trustworthiness

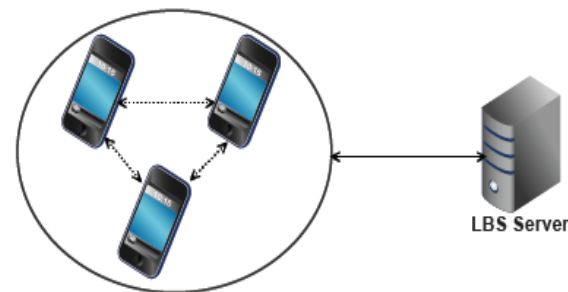
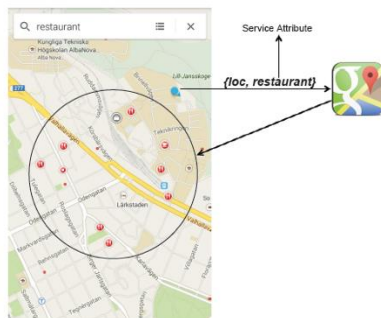
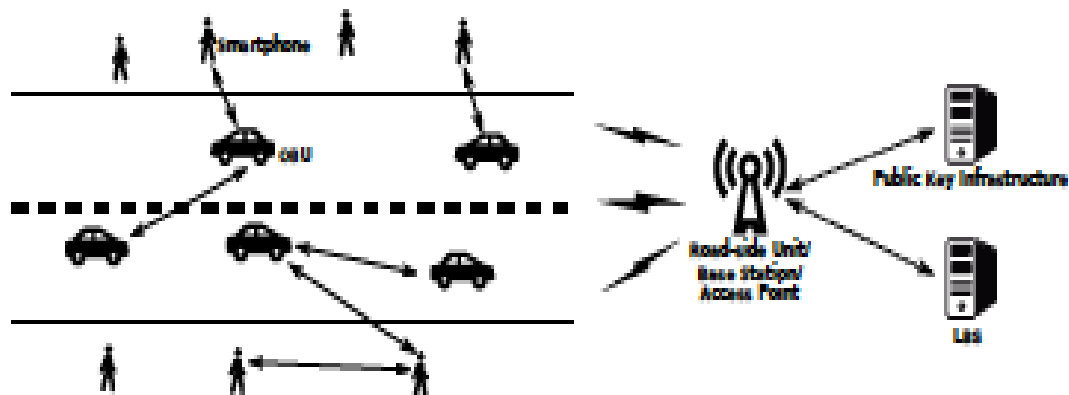
- Assess data trustworthiness
- Filter out faulty data
- Adapt to changing conditions

- **Attacks on Google Maps (2021)**
 - “Sybil-Based Attacks on Google Maps or How to Forge the Image of City Life”



Location Privacy

- Secure and Privacy-Preserving Location Based Services (LBS)
 - Challenges
 - > *Honest-but-curious service providers*
 - > *Malicious users/peers*
 - Solution
 - > *Peer-to-peer operation*
 - > *Privacy; reduced exposure to LBS servers and peers*
 - > *Security*
 - > *Efficiency*

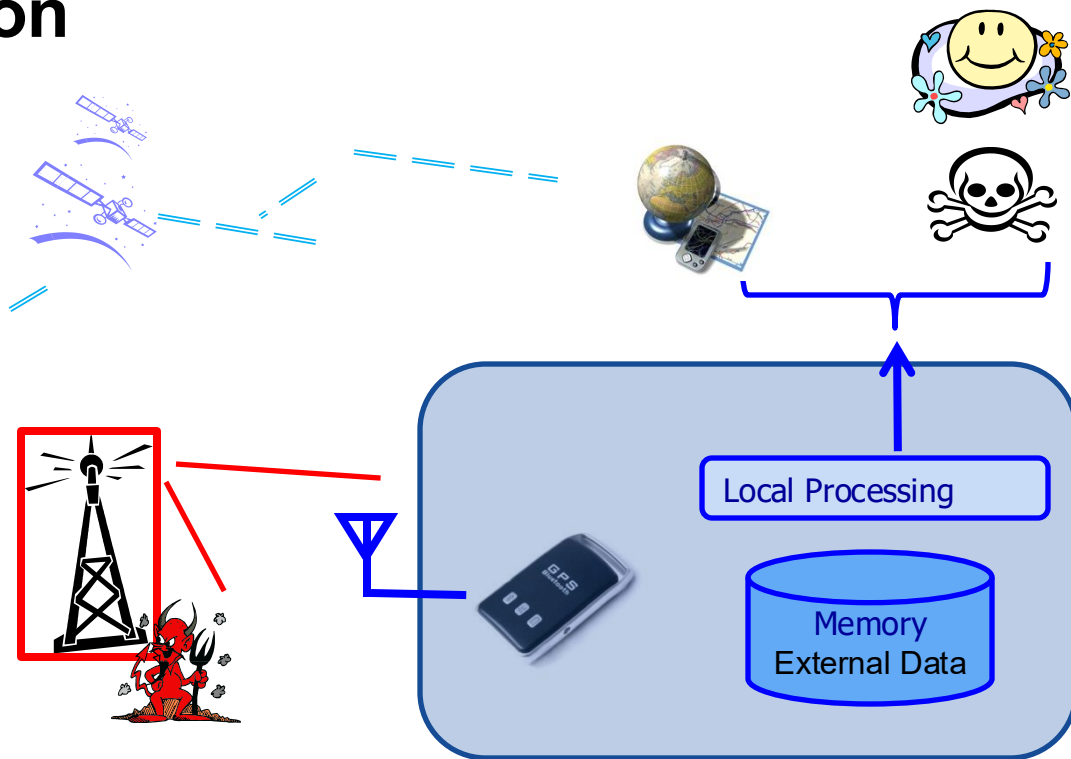


Secure localization

- Challenge
 - Attackers can easily ‘overwrite’ legitimate Global Navigation Satellite Systems
 - Victim receivers with false position and/or time
- Solution
 - Detect the attack
 - Compute a trustworthy position and clock correction

Projects with:

- European Space Agency (ESA)
- FOI



[USPO 2012, ICL-GNSS, IS-GNSS 15, IEEE Aero 2019, ION ITM 2019, ION GNSS+ 2019, ION/IEEE PLANS 2020, ION GNSS+ 2020, IEEE OJCOM 2021, IEEE TAES 2021]

Secure localization (cont'd)

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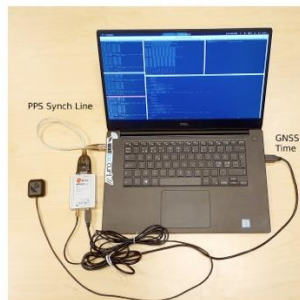
Instance of ‘late-blooming’ work
(first papers in 2008)



(a) Calculated positions with errors in Fig. 11.



(b) Corrected positions.



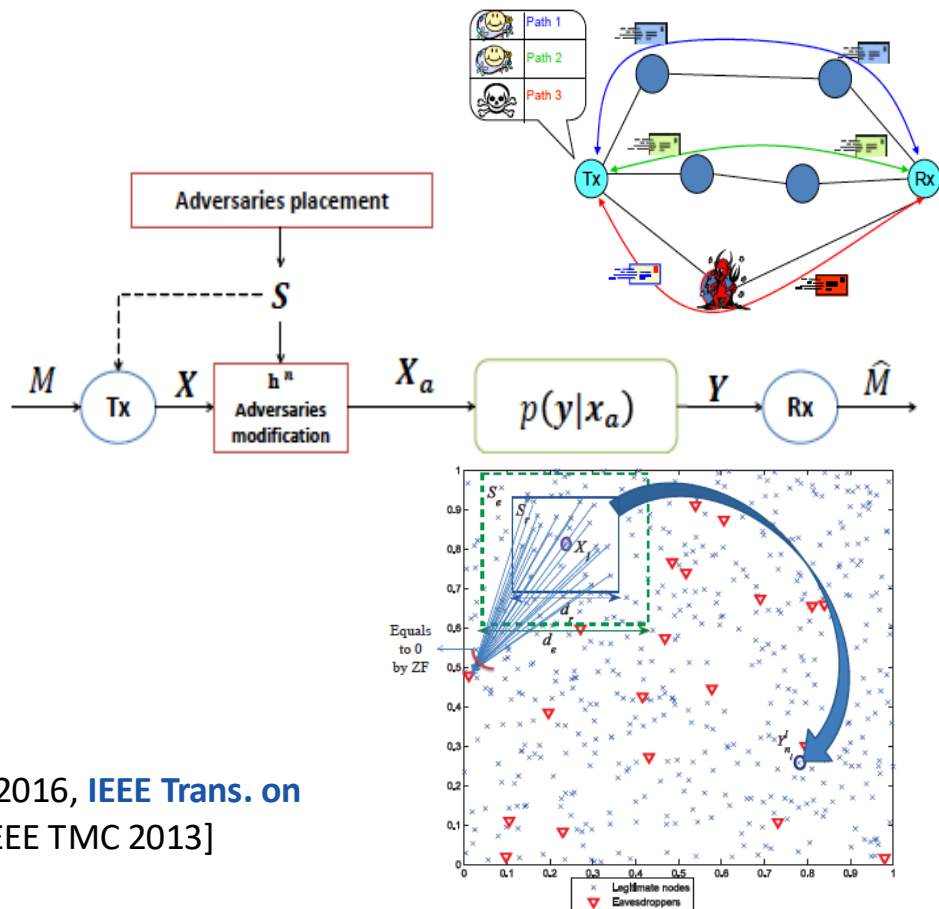
[USPO 2012, ICL-GNSS, IS-GNSS 15, IEEE Aero 2019, **ION ITM 2019**, **ION GNSS+ 2019**, **ION/IEEE PLANS 2020**, **ION GNSS+ 2020**, IEEE OJCOM 2021, **ACM WiSec 2021**, **ION ITM 2022**, **IEEE Aero 2022**]



Information-theoretic and physical layer security

- Capacity bounds against active adversaries: new IT model
- Scaling laws: Secrecy capacity: Secure communication for free as the network grows
 - “Secure Wireless Communications: Bridging theory and practice,” ESORICS 2016 keynote
 - “Secure Communication in Wireless Networks,” ACM MobiHoc Wksp. on Frontiers of Networks: Theory and Algorithms, 2016 invited talk
- Secure ranging/distance bounding
 - Another instance of ‘late-blooming’ results
- Security for battery-free backscatter tags

[IEEE ISIT 2014, IEEE INFOCOM 2014, Allerton 2015, IEEE ISIT 2016, IEEE Trans. on Information Theory 2017] [IEEE TWC 2012, IEEE TDSC 2013, IEEE TMC 2013]
[IEEE ICCCN 2021, ACM HotNets 2021]



Secure routing

Input
 $S, D \square N$

Secure Routing
Protocol



Output

An (S, D) -route

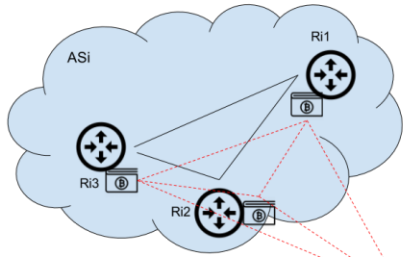
and

(i) Explicit:

$m_{0,1}, m_{1,2}, \dots, m_{n-1,n}$

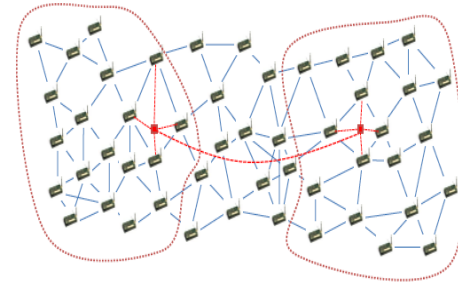
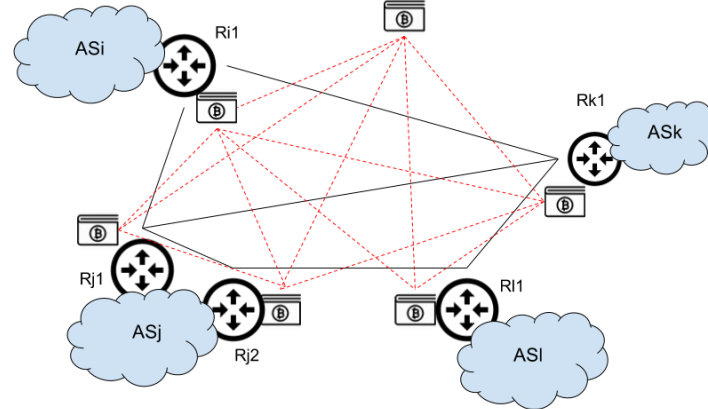
(ii) Implicit:

$g(m_{0,1}, m_{1,2}, \dots, m_{n-1,n})$



The blockchain

AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i
AS_id	operation_type	IPs	gamma	mu	i



- Protect the **route discovery**, the discovery of communication paths
 - Specification, secure neighbor discovery
- End to end path maintenance
- Gradual deployment, trust establishment, block-chain approaches

[ACM CCR 2017, iNetSec 2017, iNetSec 2015, IEEE TDSC 2013, IEEE INFOCOM 2010, ACM MC2R 2002, SCS CNDIS 2002]

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